1052 — Computer Engineering (Full Time) Agile Product Development (1052235214)

Part A — 1 mark(s) each

Unit 1

1. What is the advantage of Iterative enhancement model?
   a) No client involvement
   b) Quick development without testing
   c) Early delivery of a working product
   d) Detailed documentation before development
   Answer: *C*

2. A major difference between a program and a software product is:
   a) Programs are scalable
   b) Products are informal
   c) Products are rarely documented
   d) Programs are developed for specific needs
   Answer: *D*

3. Which model focuses on small, iterative cycles called sprints?
   a) Agile model
   b) Prototyping model
   c) Waterfall model
   d) Spiral model
   Answer: *A*

4. Prototyping model is most useful when:
   a) The system is small and simple
   b) Requirements are rigid
   c) Requirements are not well understood
   d) Project deadlines are strict
   Answer: *C*

5. Which characteristics makes software easy to modify?
   a) Scalability
   b) Flexibility
   c) Intangibility
   d) Reliability
   Answer: *B*

6. The spiral model focuses
   a) Sequential development
   b) Risk analysis
   c) Prototype model
   d) Iterative development
   Answer: *B*

7. Software does not wear out but it becomes obsolete due to
   a) Market changes and new requirements
   b) Code aging
   c) User Error
   d) Syntax error
   Answer: *A*
8. Who developed the Spiral model?
   a) Barry Boehm
   b) James Martin
   c) Margaret Hamilton
   d) Tim Berners-Lee
   Answer: *A*
9. Which of the following is NOT a characteristics of software?
   a) Flexibility
   b) Tangibility
   c) Maintenance
   d) Complexity
   Answer: *B*
10. Which is one benefit of applying software engineering practices?
    a) Low quality
    b) Increased error
    c) Higher cost
    d) Better maintenance and quality control
    Answer: *D*
11. Which of the following usually includes user manuals and guides?
    a) Program
    b) Software product
    c) Script
    d) Code snippet
    Answer: *B*
12. Which of the following is NOT a type of software prototype?
    a) Throwaway prototype
    b) Evolutionary Prototype
    c) Incremental prototype
    d) Final prototype
    Answer: *D*
13. How does the Spiral model handle changes in the requirements?
    a) Changes are not allowed
    b) Changes are expensive
    c) Changes are easily handled through iteration
    d) Changes are handled at the end
    Answer: *C*

14. A disadvantage of Iterative Enhancement Model is:
    a) No working software until last iteration
    b) Requires high level of client involvement
    c) Difficult to test small parts
    d) May require more time and resources due to rework
    Answer: *D*

15. In the Waterfall model, when testing phase is conducted?
    a) Before design
    b) Simultaneously with development
    c) After coding is complete
    d) At the beginning of the Project
    Answer: *C*

16. Which of the following is an Agile Methodology?
    a) Scrum
    b) Waterfall
    c) Big Bang
    d) Prototyping
    Answer: *A*

17. What is the primary use of a software product?
    a) Education purpose
    b) Solving individual problems
    c) General and wide-scale use
    d) For testing software
    Answer: *C*

18. In Agile, a short time-boxed development phase is called a :
    a) Stage
    b) Sprint
    c) Loop
    d) Phase
    Answer: *B*

19. Software Engineering is defined as the application of _____ approach to the development of software.
    a) Random
    b) Artistic
    c) Systematic and disciplined
    d) Informal
    Answer: *C*

20. In the Waterfall model, a change in requirements _____
    a) is easy to change
    b) is difficult and expensive
    c) is done in the next sprint
    d) has no effect

Answer: *B*

Unit 2

1. Which of the following is NOT one of the four Agile values?
   a) Customer collaboration over contract negotiation
   b) Rigid planning over adaptability
   c) Individuals and interactions over processes and tools
   d) Responding to change over following a plan
   Answer: *B*
2. DSDM STANDS FOR:
   a) Direct System Development Method
   b) Dynamic Software Development Method
   c) Distributed Software Delivery Model
   d) Design Specification Development Model
   Answer: *B*
3. Agile teams are usually:
   a) Rigidly structured
   b) Supervised by only project managers
   c) Cross-functional and self-organising
   d) Supervised by roles and departments
   Answer: *C*
4. Which is NOT considered a key in Agile Capability?
   a) Rapid delivery of working software
   b) Welcoming changes during development
   c) Continuous stakeholder involvement
   d) Comprehensive initial documentation
   Answer: *D*
5. According to Agile principles, what is the primary measure of progress?
   a) Number of completed tasks
   b) Client Satisfaction
   c) Working Software
   d) Amount of documentation
   Answer: *C*
6. Respect in Agile team means:
   a) Agreeing with everyone's opinion
   b) Ignoring junior team members' suggestions
   c) Never challenging others' ideas
   d) Valuing each member's contribution regardless of role
   Answer: *D*
7. Agile software development mainly promotes
   a) Big upfront design
   b) Iterative and incremental development
   c) Rigid documentation

d) Minimal testing

Answer: *B*

8. In Agile, the list of all desired project features is called:
   a) Sprint Backlog
   b) Product Backlog
   c) Requirements Document
   d) Feature Stack
   Answer: *B*
9. Scrum is based on which type of development cycle?
   a) Continuous
   b) Spiral
   c) Waterfall phases
   d) Time-boxed iterations (Sprints)
   Answer: *D*
10. Which statement is false about Agile teams?
    a) They include members from different functional areas
    b) They rely on one-way communication from the manager
    c) They collaborate to make decisions
    d) They aim to improve their process in every sprint
    Answer: *B*
11. A key driver of Agile adoption in modern organization is:
    a) Rapidly changing customer requirements
    b) less reliance on skilled developers
    c) Waterfall's success in complex projects
    d) The need for fixed scope in early in the project
    Answer: *A*
12. What is the purpose of a daily stand-up meeting in Agile projects?
    a) To discuss blockers and plan the day
    b) To deliver weekly reports
    c) To finalize the project scope
    d) To assign new team members
    Answer: *A*
13. Agile promotes ethical behaviour by encouraging:
    a) Open communication and mutual respect
    b) Competition between team members
    c) Strict hierarchy
    d) Avoidance of client involvement
    Answer: *A*
14. The Agile Manifesto was introduced in:
    a) 1995
    b) 2001
    c) 2007

d) 2010

Answer: *B*

15. How many core values are defined in the Agile Manifesto?
    a) 12
16. Which of the following is TRUE about Agile Documentation?
    a) It is avoided entirely
    b) It is light weight and created just in time
    c) It is created only at the end of the project
    d) It is generated automatically by the testing team

    Answer: *B*
17. In the Traditional model , software is delivered:
    a) At the end of the project
    b) After each sprint
    c) Continuously during the process
    d) Before testing

    Answer: *A*
18. Agile teams are capable of:
    a) Operating without planning
    b) Ignoring end-user needs
    c) Iterative development with continuous feedback
    d) Delivering the entire product at once

    Answer: *C*
19. Which Agile artifact contains lightweight documentation of user needs?
    a) Test Plan
    b) Sprint Backlog
    c) User Story
    d) UML diagram

    Answer: *C*
20. In Agile, how are the team members expected to interact?
    a) Through formal written reports
    b) With minimal interaction
    c) Through regular, face-to-face communication
    d) Only through the Scrum Master

    Answer: *C*

Unit 3

1. In Scrum, who has the authority to cancel a Sprint?
    a) Development Team
    b) Product Owner
    c) Scrum Master
    d) Project Manager

    Answer: *B*
2. The "Iterate" phase of Adaptive Software Development defines:

a) Specifying the Product's final design
b) Without feedback completing the development cycle.
c) Testing the product after its final release
d) Developing and refining features through short iterations based on feedback

Answer: *D*

3. How long is a typical Scrum Sprint?
   a) 1 to 4 weeks
   b) 1 day
   c) 1 to 6 months
   d) Until the project is complete

   Answer: *A*

4. Crystal Methodology encourages "frequent workshop" to:
   a) Check if the client changed their budget
   b) Train new employees
   c) Finalize documentaion
   d) Improve team process and communication

   Answer: *D*

5. What distinguishes FDD from other Agile methods like Scrum or XP?
   a) It uses burn-down charts
   b) It focuses on pair programming
   c) It is model-driven ,design- first approach
   d) It uses Kanban boards

   Answer: *C*

6. Adaptive Software Development (ASD) is best suited for:
   a) Requirements are rapidly changing and are unclear
   b) Projects with fixed requirements
   c) Large projects
   d) Projects with long-term timeline

   Answer: *A*

7. The role played by the Customer during development iterations:
   a) Only reviews the final release
   b) Manages budget and hiring
   c) Collaborates with the team daily and prioritizes user stories
   d) Design the system architecture

   Answer: *C*

8. What is the primary goal of Lean production?
   a) Increase inventory levels
   b) Maximise machine downtime
   c) Increase number of workers
   d) Reduce waste and improve efficiency

   Answer: *D*

9. What is the Extreme programming (XP) practice of "Refactoring"?

a) Rewriting entire modules from scratch
b) Improving code structure without changing functionality
c) Adding new features quickly
d) Removing unit tests to simplify code
Answer: *B*

10. The colour code of the Crystal methodologies indicate?
a) Programming languages used
b) The Client's budget
c) The number of developers and criticality of the Project.
d) The number of sprints planned
Answer: *C*

11. The core practice of Extreme Programming is:
a) Waterfall design
b) Use Case Modeling
c) Critical Path Analysis
d) Pair Programming
Answer: *D*

12. What is meant by "Test-Driven Development" in XP?
a) Testing is done after deployment
b) Tests are written before the actual code
c) Tests are written after code is completed
d) Customers write all tests
Answer: *B*

13. Adaptive Software Development focuses on:
a) Reducing the number of iterations in the project
b) Adhering to pre-defined timeline
c) Flexibility to accommodate change throughout the project life cycle
d) Detailed upfront planning
Answer: *C*

14. "5S" in Lean production stands for:
a) Simplify, Streamline,Standardise,Sustain,Speed
b) Sort,Satisfy,Standardise,Set,Safety
c) Sort,Set in order,Shine, Standardise, Sustain
d) Sort,Set up,Simplify,Standardise,Satisfy
Answer: *C*

15. How many core processes does Feature driven Development define?

16. Which of the following is NOT one of the FDD processes?
a) Develop an overall model
b) Build a feature backlog
c) Design by feature
d) Plan by feature
Answer: *B*

17. Which Crystal methodology is designed for largest and more critical projects?
    a) Crystal Clear
    b) Crystal Red
    c) Crystal Maroon
    d) Crystal Orange
    Answer: *C*
18. Which of the following is a primary work product in Extreme Programming?
    a) User stories
    b) Functional specification documents
    c) UML diagrams
    d) Business case studies
    Answer: *A*
19. What is a challenge in implementing Adaptive Software Developmet (ASD)?
    a) High upfront documentation
    b) Lack of flexibility in scope
    c) Need for high communication and collaboration
    d) Following a rigid development process
    Answer: *C*
20. Which is NOT a Scrum artifact?
    a) Product Backlog
    b) Gantt Chart
    c) Increment
    d) Sprint Backlog
    Answer: *B*

Unit 4

1. Scrum Master's role in decision making is to:
    a) Control the development team
    b) Facilitate team collaboration and remove obstacles
    c) Make final business decisions
    d) Enforce decisions made by the managers
    Answer: *B*
2. Applying Institutional knowledge to solve real-world problem is called _____stage of knowledge evolution cycle.
    a) Knowledge Retention
    b) Knowledge Application
    c) Knowledge Sharing
    d) Knowledge Development
    Answer: *B*
3. The Systems school focuses on:
    a) Organisational story telling
    b) Revenue Generation through patents
    c) Knowledge embedded in IT systems and databases

d) Human resource strategies

Answer: *C*

4. In the widely recognized SECI model of knowledge management, which process is specifically defined as the conversion of tacit knowledge into explicit knowledge?
   a) Socialization
   b) Externalization
   c) Combination
   d) Internalization
   Answer: *B*
5. What is the first stage of the Institutional Knowledge Evolution Cycle?
   a) Knowledge Sharing
   b) Knowledge Development
   c) Knowledge Acquisition
   d) Knowledge retension
   Answer: *B*
6. Which KM process is involved when a senior developer teaches junior developer through pair programming?
   a) Knowledge retention
   b) Knowledge Application
   c) Knowledge acquisition
   d) Knowledge Transfer
   Answer: *D*
7. Who proposed the "Schools of Knowledge Management" framework?
   a) Peter Drucker
   b) Barry Bohem
   c) Karl Wiig
   d) Michael Earl
   Answer: *D*
8. Name of the tool that supports both verson control and collaborative knowledge sharing is:
   a) Github
   b) Slack
   c) Excel
   d) Paint
   Answer: *A*
9. The Story Card Maturity Model is used during:
   a) Backlog grooming and sprint planning
   b) Sprint review
   c) Product demo
   d) Sprint retrospective
   Answer: *A*
10. Which Agile artifact supports continuous knowledge sharing within the team?
    a) Daily Scrum Meeting

b) Product backlog
c) Burn-up chart
d) Definition of Done
Answer: *A*

11. A lesson learned repository in a software company is an example of:
a) Informal communication
b) Knowledge destruction
c) Tacit knowledge
d) Explicit knowledge
Answer: *D*

12. Which describes an Agile Information Systems?
a) A system used in Government sectors
b) A rigid system with fixed work flows
c) A system that requires complete requirements before development
d) A flexible,user-centered system that evolves with business needs
Answer: *D*

13. In Agile, faster decision-making leads to:
a) Quicker delivery and adaptability
b) Less team autonomy
c) More documentation overhead
d) Reduced communication
Answer: *A*

14. Which describes a level 5 (Ready) story in SMM?
a) Waiting for product owner review
b) Has no technical alignment
c) Complete, small, testable and ready for implementation
d) Contains unresolved dependencies
Answer: *C*

15. What is a key difference between Level 3 and Level 5 in the SMM?
a) Level 3 stories are larger than epics
b) Level 5 stories are immediately actionable and testable
c) Level 5 stories are no longer discussed
d) Level 3 includes testing steps
Answer: *B*

16. What is the primary characteristic of decision-making in Agile teams?
a) Based on contracts
b) Centralised and Hierarchial
c) Decentralised and collaborative
d) Delayed until project completion
Answer: *C*

17. The Name of the tool that is not used for knowledge management in Software Engineering
a) GitHub

b) Confluence
c) Microsoft Paint
d) Stack Overflow
Answer: *C*

18. A Sprint retrospective is conducted to:
    a) Assess team performance and identify improvements
    b) Measure code length
    c) Release final product
    d) Review customer contracts
    Answer: *A*
19. How many levels are commonly used in the Story card Maturity Model?
    a) 10
20. In Software engineering, which activity supports knowledge reuse?
    a) Using version control only for backups
    b) Limiting team collaboration
    c) Maintaining code libraries and design patterns
    d) Writing new code for every project
    Answer: *C*

Unit 5

1. In Agile, when are the requirements finalized?
    a) At the beginning of the project
    b) After Sprint Retrospective
    c) During Sprint Planning
    d) Never completely finalized-they evolve continuously
    Answer: *D*
2. The Agile practice that helps in validating stakeholder's requirements regularly
    a) Sprint Review
    b) Sprint Retrospective
    c) Stand-up Meeting
    d) Backlog Grooming
    Answer: *A*
3. How does Agile Requirement Engineering differ from Traditional Requirement Engineeing?
    a) Agile RE requires a completed Software Requirement Specification (SRS) before development
    b) Agile RE focuses on adaptabilty and minimal upfront requirements
    c) Agile RE is more formal and structured
    d) Agile RE avoids documentation
    Answer: *B*
4. The amount of work , a team completes during a sprint is called:
    a) Lead Time
    b) Velocity
    c) Cycle Time
    d) Burndown Rate

Answer: *B*

5. The technique used for prioritizing requirements in Agile is:
   a) Data Flow Diagram (DFD)
   b) Class Diagram
   c) MoSCoW Method
   d) Data Dictionary
   Answer: *C*
6. What is the primary goal of Requirements Engineering in Agile?
   a) Freeze requirements before development
   b) Capturing changing stakeholder needs
   c) Maximize documentation
   d) Create detailed documents upfront
   Answer: *B*
7. What Agile practice manages unstable requirements effectively?
   a) User story mapping and backlog grooming
   b) Requirement freezing
   c) Creating a SRS document
   d) Daily stand-up meetings
   Answer: *A*
8. Which of the following is a key reason for unstable requirements?
   a) Lack of internet connection
   b) Changing business needs or stakeholder priorities
   c) Overqualified development team
   d) Too much documentation
   Answer: *B*
9. When does Quality Assurance (QA) start?
   a) At the beginning and throughout the Project
   b) After development is complete
   c) During the final sprint
   d) After Product release
   Answer: *A*
10. What is meant by "unstable requirements" in Software Engineering?
    a) Requirements that are unclear due to poor handwriting
    b) Requirements that are too technical
    c) Requirements that are not documented
    d) Requirements that frequently changes over time
    Answer: *D*
11. What does the "MoSCoW" technique stand for in requirements prioritization?
    a) More, Some, Critical, Worthy
    b) Mandatory, Optional, Suggested, Canceled
    c) Model, System, Cost, Workflow
    d) Must have, Should have , Could have, Won't have

Answer: *D*

12. The Agile ceremony that helps to refine and manage requirements.
    a) Sprint Retrospective
    b) Backlog Refinement (Grooming)
    c) Daily Scrum
    d) Sprint Review
    Answer: *B*

13. What is the main purpose of prioritizing requirements in Agile development?
    a) To reduce developer workload
    b) To delay unnecessary work
    c) To deliver the highest business value first
    d) To eliminate less important stakeholders
    Answer: *C*

14. What is the advantage of using workshops in elicitation?
    a) No need for user input
    b) Allows group discussions and consensus building
    c) Reduces Project budget
    d) Faster coding
    Answer: *B*

15. In Agile, how are requirements primarily managed?
    a) Using UML diagrams
    b) Through Waterfall-based documentation
    c) using a dynamic Product Backlog
    d) Through a complete upfront SRS document
    Answer: *C*

16. Which of the following is NOT a technique used for requirements elicitation?
    a) Observation
    b) Brainstorming
    c) Coding
    d) Interviews
    Answer: *C*

17. What is the primary goal of requirements elicitation?
    a) Designing the software Architecture
    b) Testing the system
    c) Writing source code
    d) Understanding stakeholder needs
    Answer: *D*

18. In Agile RE, what is the typical format of user story?
    a) "System shall perform [function]"
    b) "Do this, then that"
    c) "As a [user], I want [feature] so that [benefit]"
    d) "Given, When, Then"

Answer: *C*

19. In which format, Agile requirements are typically captured?
    a) Use case diagrams
    b) Formal specification documents
    c) Entity- Relationship diagram
    d) User stories
    Answer: *D*

20. Who is primarily responsible for managing the product backlog in Agile?
    a) Product Owner
    b) Scrum Master
    c) Developer
    d) Project Manager
    Answer: *A*

Part B — 2 mark(s) each

Unit 1

1. Define Program and Software Product.
   Answer: *1.Program is a set of instructions to do a specific job*
   It is a combination of Source code and Object code.
   2.Software product consists of Program code and all associated documents like
   requirements specification documents,the design documents ,the test documents and operating procedures.

2. List any four needs for Software Engineering.
   Answer: *1.Large Software 2. Modularity 3.Scalability 4.Incremental Development*

3. Define Software Engineering.
   Answer: **Systematic approach to Software Development.*
   **IEEE standard definition*
   Software Engineering is the application of systematic,disciplined,quantifiable approach to the development, operation and maintenance of Software.

4. Write the features of Waterfall Model.
   Answer: *The Waterfall model is a linear, sequential approach in software development, where each phase must be completed before the next can begin, much like water flowing over a cliff.*

5. Write the main advantage of Spiral model.
   Answer: *It helps in identifying and reducing risks early in the development process.*

6. What is Prototyping Model?
   Answer: *It is a Software Development model where a prototype is built,tested and refined based on user feedback before final development begins.*

7. List the different phases of Software Development Life Cycle (SDLC) model.
   Answer: *1.Problem definition*
   2.Project identification / Feasibility Study
   3.Requirement Analysis
   4.Design

5.Coding 6.Testing 7.Implementation and Maintenance

8. Write the features of Iterative enhancement model.
   Answer: *(i) It is the combination of Linear Sequential Model and Iterative philosophy of prototyping model.*
   (ii) The development team develops the core module first.
   (iii) Then later it is refined and new functionalities are added in the successive version.
9. What is the Agile model?
   Answer: ** It is a combination of Iterative and Incremental model.*
   ** Focusing on process adapting and customer satisfaction.*
   ** Rapid delivery of working software product.*
10. List any four Software characteristics.
    Answer: *1. Most software is custom built.*
    2.Software is developed not manufactured.
    3.Software is flexible
    4.Software does not wear out.

Unit 2

1. List any four Agile drivers.
   Answer:
   Change
   Customer Demands
   Market Competition
   Technological Advancements
   Globalization
2. Give examples for Agile documentations.
   Answer:
   Product Backlog
   User Stories
   Burn-down and Burn-up Charts
   Technical Documentation
3. Write about Agile Software Development.
   Answer:
   ** It is a Software Development* ***Methodology.***
   **It values flexibility, collaboration, and customer satisfaction.*
   ** It is an* ***iterative and incremental approach*** *to Software Development****.***
4. What is an Agile team?
   Answer:
   A collection of people arranged to collaborate effectively and deliver useful goods or services is called an agile team.
5. List the phases of Agile Project Management.
   Answer:
   1. Envision 2. Speculate3. Explore 4. Adapt 5. Close

6. List the ethical considerations in Agile Teams.

Answer:

Data Privacy

Human Value Preservation

Bias and Fairness

Accessibility and Inclusion

Ethical Design

7. List any two differences between Traditional model and Agile model.

Answer: *Traditional Model*

**It is used to develop simple software*

**Testing is done once the development phase is compeleted*

Agile Model

**It is used to develop complicated software.*

**Testing and development processes are performed concurrently.*

8. List the major classification of Agile methods.

Answer:

·Agile methods can be broadly classified into frameworks and methodologies.

·Frameworks like Scrum and Kanban provide a structured approach, while methodologies like Extreme Programming (XP) and Feature-Driven Development (FDD) offer more specific practices.

·Additionally, approaches like Lean Software Development focus on principles rather than specific frameworks.

9. List the four Agile values.

Answer:

Individuals and Interactions over Processes and Tools Working Software over Comprehensive Documentation Customer Collaboration over Contract Negotiation Responding to Change over Following a Plan

10. Write the concepts of Agile drivers, capabilities and values.

Answer:

Agile drivers create the need for agility.

Agile capabilities enable organizations to respond effectively.

Agile values guide the way they operate to achieve agility.

11. Write note on any two Agile capabilities.

Answer: Flexibility/Adaptability:

The capacity to adjust plans, processes, and strategies in response to changing circumstances.

Speed:

The ability to deliver value quickly and efficiently, often through iterative development and continuous delivery.

Unit 3

1. What is meant by lean production?

Answer:

Lean production is also known as lean manufacturing. It is a production philosophy focused on minimizing waste and maximizing productivity within a manufacturing system.

2. What is the work product in Extreme Programming (XP)?

   Answer:

   Code, test, user-stories, documentation and working software.

3. What makes Crystal different from other Agile methods?

   Answer: *Crystal is a multiple methodology ,each tailored to project's specfic characteristics like team size and criticality with a core focus on people, communication and adaptability.*

4. What are three main phases of Adaptive Software Development (ASD)?

   Answer: *(i) Speculate*
   (ii)Collaborate
   (iii)Learn

5. What is pair programming in Extreme Programming (XP)?

   Answer:

   Pair programming is an Agile software development technique where two developers work together on a single workstation to write code. One programmer, the driver, writes code while the other, the navigator, reviews the code in real-time, offers guidance, and discusses the overall direction of the task. It improves code quality and knowledge sharing.

6. What is sprint in Scrum?

   Answer: *It is a time box period to complete the work. It is between 1 week to 4 weeks period.*

7. Name any two practices of Extreme Programming (XP)?

   Answer: *(i) Pair –Programming (ii) Test Driven Development (TDD)*

8. Name two key roles in scrum team.

   Answer:

   Product Owner: Defines the product vision and manages the
   product backlog.
   Scrum Master: Facilitates the Scrum process and helps the team
   optimize their workflow.

9. Name two techniques used in Lean production?

   Answer: *Two key techniques used in lean production are Kaizen (Continuous Improvement) and Just-in-Time (JIT) production.*

10. Name the five basic activities of Feature Driven Development.

    Answer: *·Build overall model*
    Build feature list
    Plan by feature
    Design by feature
    Build by feature

Unit 4

1. Name the classification in Knowledge Management.

   Answer: Tacit Knowledge:

   Deeply personal, often acquired through experience, and difficult to articulate or document.

   Explicit Knowledge:

   Structured, documented, and easily shared, such as code, manuals, and formal processes.

   Implicit Knowledge:

Practical application, patterns, and insights derived from experience that may not be fully codified but are shared through social interaction.

2. What is Institutional Knowledge Evolution Cycle?
   Answer: *The institutional knowledge evolution cycle outlines how organizations develop, manage, and leverage collective knowledge to improve their software development processes.*
3. What is Agile Information System?
   Answer: *Agile Information Systems (AIS) refer to information systems designed, developed and maintained using agile methodologies to deliver flexible, adaptive, and user-focused solutions*
4. What does a Story card Maturity Model represent (SMM)?
   Answer: *The Story Card Maturity Model (SMM**)** is a conceptual framework used in Agile software development to assess and improve the quality and readiness of user stories before and during a sprint. It helps the teams to ensure that each story card is clear, complete and ready for implementation. It defines the progressive levels of maturity of user stories.*
5. What are the key stages in Institutional Knowledge Evolution Cycle?
   Answer: *(1) Knowledge development (2)Knowledge Acquisition (3)Knowledge refinement*
   (4) Knowledge Distribution (5) Knowledge Deployment (6) Knowledge Leveraging
6. What is meant by a "mature" story card?
   Answer: *A mature story card is well-defined, understood by the team, testable, and ready for development without blockers.*
7. Name any two schools from Earl-s Knowledge Management framework.
   Answer: *Systems School*
   It emphasises codified knowledge stored in systems, often with validation from peers.
   Commercial School
   It treats knowledge as as asset to generate revenue.
8. Name the three broad categories under which Earl classified the KM schools.
   Answer: *The three categories are (i) Technocratic (2) Economic (3) Behavioural*
9. Give two examples for Knowledge management tool used in Software Engineering.
   Answer: Knowledge Management Systems (KMS):
   Platforms like Confluence, SharePoint, or dedicated KM software provide a centralized repository for storing and sharing knowledge.
   Wikis:
   Internal wikis can be used to create and maintain documentation and knowledge bases.
   Version Control Systems:
   Git and other version control systems are essential for managing code and related documentation.
   Collaboration Tools:
   Tools like Slack, Microsoft Teams, or dedicated project management platforms facilitate communication and knowledge sharing.
10. What is Agile Decision Making?
    Answer:
    Agile Decision Making (ADM) is a dynamic, collaborative, and iterative approach to make decisions in fast-paced, uncertain or complex environments.

Unit 5

1. Write the advantages of using Agile in Requirement Engineering.

   Answer:

   Advantages

   Better alignment *with business needs.*

   Reduced risk *of delivering irrelevant features.*

   Improved quality *through continuous validation.*

   Encourages ***innovation*** *by allowing late requirement changes.*

2. List the steps of Requirements Elicitation.

   Answer: *Steps of Requirements Elicitation*

   1.Identify Stakeholders

   2.Gather Requirements

   3.Prioritize Requirements

   4.Categorize Feasibility

3. Why requirements are unstable?

   Answer:

   In projects, requirements change because:

   Market changes *(competitors, trends, regulations).****Stakeholder feedback*** *after seeing working software.****Technical discovery*** *during development.****Evolving user needs****.*

4. What is Requirement Engineering?

   Answer:

   Requirements Engineering (RE) is the structured process of identifying, documenting, analyzing, and managing the needs and expectations of stakeholders for a system or software product.

5. Write the challenges of using Agile in Requirement Engineering.

   Answer:

   Challenges

   Scope creep risk *due to continuous changes.****Harder to predict*** *long-term budgets and timelines. Requires* ***high customer availability*** *and strong communication. Teams need* ***discipline*** *to maintain a minimal but sufficient level of documentation.*

6. Write the challenges in Agile requirements Management.

   Answer:

   Challenges in Agile Requirements Management

   Scope creep if backlog is not tightly prioritized. Requires strong Product Owner discipline. Balancing flexibility with commitment can be tricky. Frequent changes can affect team velocity predictions.

7. Write the Agile Requirements Management Process.

   Answer:

   Agile Requirements Management Process

   Capture *– Initial requirements from stakeholders.****Prioritize*** *– Order backlog by value, urgency, and risk.****Refine*** *– Break down and clarify items before each sprint.****Plan*** *– Select sprint backlog during sprint planning.****Implement & Test*** *– Build and verify in short iterations.****Review & Adapt*** *– Gather feedback and update backlog.****Repeat*** *– Continuous cycle until goals are met.*

8. Why Requirements prioritization is important in Agile?

   Answer:

Importance of Requirements Prioritization in Agile

Delivers maximum value early. Makes the most of limited time and budget. Adapts to changing requirements without chaos. Helps stakeholders make trade-offs between features.

9. What is Requirements Elicitation? Why is it important?

Answer:

Requirements Elicitation

The process of investigating and learning about a system's requirements from users, clients, and other stakeholders is known as requirements elicitation.

Requirements elicitation is most difficult, most error-prone and most communication-intensive software development in Software Engineering.

10. How Agile changes in Requirement Engineering?

Answer:

In Agile:

Requirements are iteratively discovered, refined, and reprioritized throughout the project. They are captured in lightweight formats like user stories, epics, and acceptance criteria. Customer collaboration is prioritized over contract negotiation.

11. Write the benefits of using Agile in case of unstable requirements.

Answer:

Benefits of Agile for Unstable Requirements

Lower rework costs *(because work is done in small increments).****Better alignment*** *with current business priorities.****Higher customer satisfaction*** *because delivered features reflect up-to-date needs.*

Part C — 10 mark(s) each

Unit 1

1. Draw and explain the features of Spiral Model.

Answer:

The spiral model is a risk-driven Software Development Life Cycle (SDLC) model that combines iterative development with the systematic control of a sequential model, progressing through cycles of planning, risk analysis, engineering, and evaluation. It is particularly suited for large, complex, and expensive projects where risks are a major concern, as it allows for continuous risk assessment and mitigation in each iteration or "spiral" of the project.

Core Concepts

Iterative and Incremental:

The model repeats the development process in successive cycles, with each iteration building upon the previous one, leading to increasingly refined versions of the software.

Risk-Driven:

A central focus is on identifying, analyzing, and managing risks at each stage of the project.

Phased Approach:

Each complete iteration, or round, consists of four main phases:

Identification (or Planning): *Gathering business requirements and defining the project's goals.* ***Risk Analysis:*** *Identifying and resolving project and technical risks, such as cost overruns or technical feasibility issues.* ***Engineering (or Development):*** *Developing and constructing the software.* ***Evaluation (or Customer Evaluation):*** *Testing and evaluating the completed software with the customer to gather feedback and determine the next steps.*

Advantages

Strong Risk Management: *Excellent for managing risks in large and complex projects.* ***Adaptability:*** *Allows for flexibility and adaptation to changing requirements.* ***Early Feedback:*** *Facilitates customer feedback, leading to a better-understood product.*

Disadvantages

High Cost: *Can be expensive, making it unsuitable for small projects.****Complexity:*** *The model is complex, requiring expertise in risk assessment to be managed effectively.****Documentation:*** *Requires significant documentation to track intermediate phases.****Time Management:*** *The unknown number of required phases can make time management challenging.*

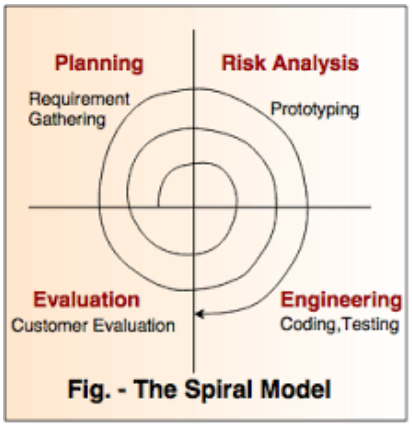


Fig. - The Spiral Model

2. Explain Program vs Software Product.

Answer: *Program is a combination of Source code and Object code.*

Software product consists of Program and all the associated documents like requirement specification documents, the design documents ,the test documents and operating procedures.

PROGRAM SOFTWARE PRODUCT

(i) Developed by individuals for their personal use. Developed by a group of engineers as a team

(ii) Small in size. Large in size.

(iii) Single user. Large number of users.

(iv) Lack of proper documentation. Good Documentation.

(v) Adhoc development Systematic development.

(vi)Lack of user interface. Good user Interface.

(vii) Have limited functionality Have more functionality.

3. Explain the need for Software Engineering and explain Software Characteristics.

Answer: *Need for Software Engineering*

(i)Large Software

Software Engineering helps to build large software.

(ii)Modularity

Software Engineering helps to develop software into components according to functionality and responsibilty.

(iii)Scalability

It helps to develop scalable software.This ability helps the software to function well when it is changed in size to meet the user need.

(iv)Incremental Development

Software Engineering helps to develop software in small increments.

(v)Cost Effective

Software Engineering helps to control the software development cost.

(vi)Quality Management

Software Engineering helps to develop quality software products.

Software Characteristics

(i) Most software is custom-bulit.

(ii) Software is developed or engineered not manufactured.

(iii)Software is flexible

(iv) Software doesnot wear out.

(v) The software is not vulnerable to external factors.

(vi)The management of software is a highly demanding task.

(vii) If the design is good, it can never be 100% error free.

4. Draw and explain the features of waterfall model.

Answer:

The waterfall model is a linear, sequential software development methodology where each phase must be completed before the next begins, mimicking a waterfall flowing downwards. It consists of distinct stages such as requirements gathering, design, implementation (coding), testing, deployment, and maintenance, with strict documentation at each step.

Characteristics

Linear and Sequential:

Each phase is completed in a strict order, one after another.

Fixed Stages:

Phases like requirements, design, implementation, testing, and maintenance are well-defined and must be completed sequentially.

Strict Documentation:

Comprehensive documentation is created at each stage for clarity and reference.

No Overlapping:

Phases do not overlap, ensuring a structured progression through the development lifecycle.

Phases of the Waterfall Model

Requirements Analysis: *Gathering and documenting all the requirements for the software.* ***Design:*** *Creating the system's design based on the gathered requirements.* ***Implementation (Coding):*** *Writing the actual code based on the design specifications.* ***Testing (Verification):*** *Verifying that the software meets the specified requirements and correcting any bugs.* ***Deployment:*** *Releasing the functional software to the user or live environment.* ***Maintenance:*** *Providing ongoing support, updates, and improvements after the software is deployed.*

Advantages

Simplicity: *The model is easy to understand and manage.****Structured:*** *It provides a structured and organized workflow.****Documentation-Heavy:*** *Thorough documentation at each stage supports future reference.*

Disadvantages

Inflexibility:

It is difficult to accommodate changes in requirements once a phase is complete.

High Risk for Complex Projects:

Not ideal for complex or long-term projects..

No Working Software Until Late:

A working version of the software is not available until much later in the development cycle.

Limited Error Detection:

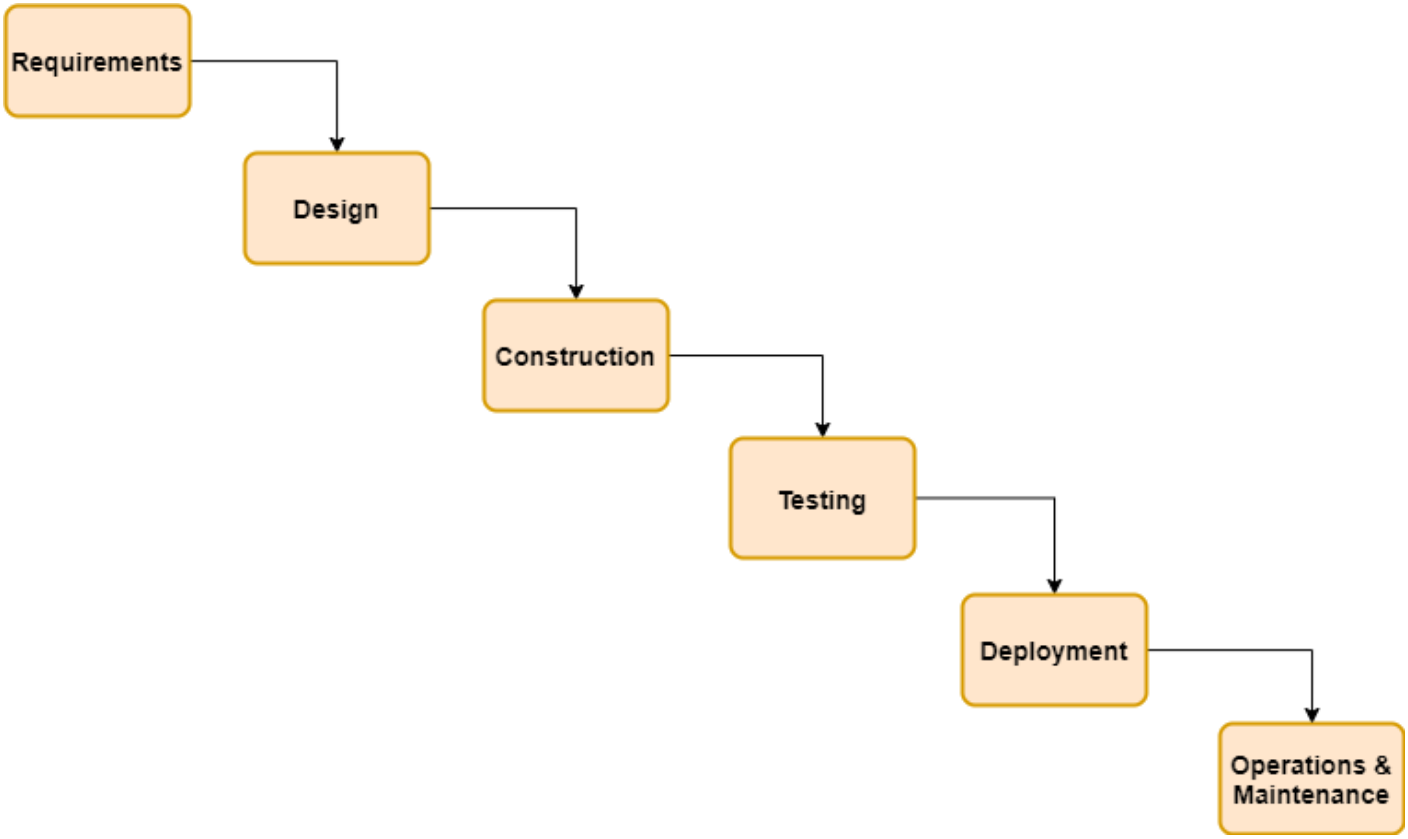
Risks and challenges are not identified until later stages, making early correction difficult.

When to Use It

The waterfall model is best suited for projects

** where requirements are well-understood, fixed, and unlikely to change*

** making it a good choice for small, simple projects with clear goals.*



Unit 2

1. Differentiate between Traditional and Agile Model.

Answer:

Traditional Software Development	***Agile Software Development***
It is used to develop simple software.	*It is used to develop complicated software.*
Testing is done once the development phase is completed.	*Testing and development processes are performed concurrently.*
It follows a linear expectation structure.	*It follows an iterative organizational structure.*
It provides less security.	*It provides high security.*
Client involvement is less.	*Client involvement is high.*
It provides less functionality.	*It provides all the functionality needed by the users.*
It supports a fixed development model.	*It supports a changeable development model.*
It is used by freshers.	*It is used by professionals.*

Traditional Software Development	***Agile Software Development***
Development cost is less.	*Development cost is high.*
It is less used by software development firms.	*It is normally used by software development firms.*
Traditional software development approaches are formal in terms of communication with customers.	*Agile software development methodologies are casual.*
For starters, software development approach is predictive approach. The time and cost of project development are fixed.	*A flexible approach is used. re Continual design improvement.*
Examples	***Examples***
Office productivity suites Data management software Media players Security programs	*Sky Phillips JP Morgan Chase*
Models based on Traditional Software Development-	*Models based on Agile Software Development-*
Spiral Model Waterfall Model V Model Incremental Model	*Scrum Extreme Programming (XP) Crystal Dynamic Systems Development Method (DSDM) Feature Driven Development (FDD) Adaptive Software Development (ASD)*

2. Explain the features of Agile documentations with examples.

Answer: AGILE DOCUMENTATIONS *Agile documentation highlights creating and maintaining just enough, just-in-time documentation. It is lightweight, collaborative and responsive to change. It prioritizes working software over comprehensive documentation. It focuses on delivering value through iterative development.*

Features:

Lightweight: *Agile documentation focuses on capturing the essential information needed to support the development process, often using concise formats like user stories, diagrams, or templates.* ***Just Enough, Just in Time:*** *Documentation is created when it's needed (just in time). This avoids creating unnecessary or outdated documentation.* ***Collaborative:*** *Agile teams collaboratively create and maintain documentation, ensuring everyone has access to the information they need.* ***Iterative and Adaptive:*** *Agile documentation evolves alongside the project.* ***Value-Driven:*** *It supports the development process and delivers value to the customer.* ***Focus on Working Software:*** *Agile principles prioritize delivering working software over extensive documentation.*

Examples of Agile Documentation:

Product Backlog*: A prioritized list of features and requirements for the product.* ***User Stories****: Brief descriptions of features from the user's perspective.* ***Burn-down and Burn-up Charts****: Visual representations of project progress.* ***Technical Documentation:*** *Limited documentation explaining code, APIs or system architecture created when needed.*

3. Draw and explain the phases of Agile Project Management.

Answer:

AGILE PROJECT MANAGEMENT (APM)

Agile Project Management (APM) highlights ***c****ollaboration, customer feedback and rapid delivery of functional components.*

It allows teams to respond quickly to changes and evolving requirements.

It enables the team members to work collaboratively with each other and with the customer.

It motivates team members to faster development of the project.

It works by breaking the processes into smaller chunks known as sprints.

It entertains customers' feedback, which helps the development team to the changes in the sprint.

Phases of Agile Project Management

1. Envision

The first phase of agile project management involves the goal and purpose of the project. It also creates a list that prioritizes the features and requirements.

2. Speculate

It involves breaking the process into smaller chunks known as sprint and detailed planning is done to how to achieve the goal. In this phase, user stories are selected from the product backlog.

3. Explore

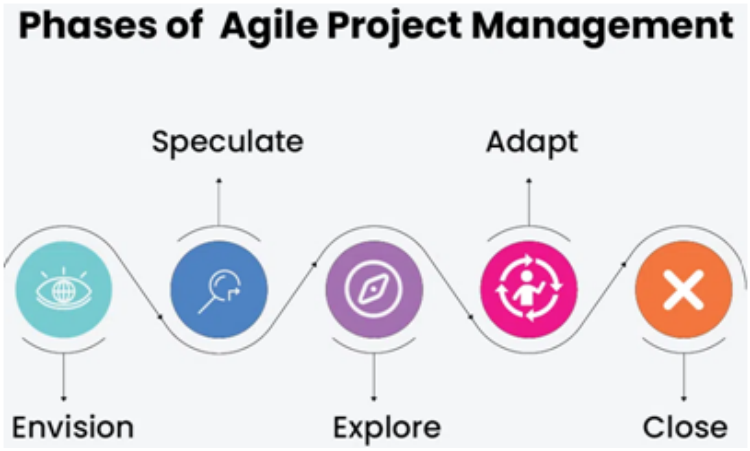
In this phase, the product is divided into sprints which last for 1-4 weeks. Progress, challenges, and plans are discussed in daily scrum meetings, and execution of the project is done.

4. Adapt

*It involves continuous adaptation of the plan based on the feedback provided by the client**.** Scrum master continues to monitor the progress of the project and also address the problems faced by the developers. It demonstrates what has happened and what needs to be done.*

5. Close

It is the last phase of agile product management. It deploys the final product or new version of the product into the real world. It collects feedback from customers for future improvement of the project



4. Explain the roles and resposibilities of an Agile team.

Answer:

Agile Team

A collection of people arranged to collaborate effectively and deliver useful goods or services is called an agile team.

Roles and Responsibilities of an Agile Team

·***Product Owner****:*

·The product owner is responsible for representing the stakeholders and the end customer.

·They define the product vision. The roadmap prioritizes the product backlog and accept or reject the outcome. They ensure the team is delivering maximum business value.

Scrum Master: *The scrum master is responsible for managing the agile processes and removing obstacles that prevent the development team from being productive. They facilitate meetings, promote collaboration, and encourage the team to improve its practices. In simple terms, they act as a binding force towards*

*reaching conclusion of the project.**Development Members / Developers:** The development team is usually structured yet flexible and self organized while carrying their task execution.**Stakeholders**:*

Stakeholders are individuals or groups who are impacted by the project.

They may be within or external to the organization.

They provide feedback on the requirements while reviewing progress of the project and also evaluate the final product.

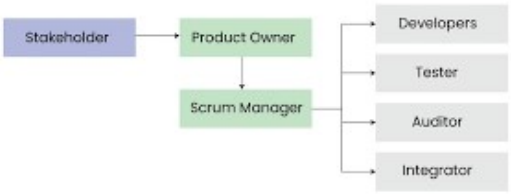
Integrator:

The integrator is responsible for technical integration of work by the development team.

They ensure that the software / hardware components fit together properly.

The integrator also makes sure the system meets quality standards and integrates smoothly with external systems.

***Independent Auditor and Tester:** The independent tester objectively evaluates the system to ensure the quality of the product. They audit the product for bugs and flaws from an unbiased perspective. The tester identifies defects and works with the team to get them fixed.*



Unit 3

1. Explain Lean Production Agile Process.

Answer:

LEAN PRODUCTION

*Lean production also known as **lean manufacturing,** is a production philosophy focused on minimizing waste and maximizing productivity within a manufacturing system.*

Principles:

***Value**:*

Identify and focus effort on what the customer truly values in a product or service.

Value Stream Mapping:

Analyzing the entire production process to identify and eliminate waste (non-value-added activities).

***Flow**:*

Ensuring a smooth, continuous flow of work through the production process, minimizes delays.

***Pull**:*

Producing goods or services only when they are needed .(demand-driven).

Perfection:

Striving continuously for improvement.

Features:

Waste Reduction:

Lean production aims to eliminate various forms of waste, including overproduction, waiting, transportation and inventory.

Just-in-Time (JIT):

JIT focuses on producing goods or services as they are needed, minimizing inventory and storage costs.

Continuous Improvement (Kaizen): *Lean production is a philosophy of continuous improvement. Employees at all levels are encouraged to identify and implement small, incremental changes to enhance processes and reduce .* ***Employee Involvement:***

Lean production highlights the importance of employee involvement and empowerment.

Benefits:

Reduced Costs:

By minimizing waste and improving efficiency lead to lower operating costs.

Increased Productivity:

Streamlined processes and optimized workflows can boost overall productivity.

Improved Quality:

Identifying and eliminating the root causes of defects can lead to higher quality products.

Faster Lead Times:

Lean production principles can help reduce the time it takes to produce and deliver products.

Greater Flexibility:

Lean systems are adaptable to changing customer needs and market demands.

2. Explain Crystal methodology of Software Development.

Answer: CRYSTAL

CRYSTAL is one of the Agile SDLC models.

The crystal method is an agile framework.

It is considered a ***lightweight*** *or agile methodology that focuses on individuals and their interactions.*

The methods are color-coded to significant risk to human life.

It is mainly for short-term projects by a team of developers working out of a single workspace.

Two core beliefs of the Crystal method:

Find your own way and methods to optimize workflow. Make use of unique methods to make the project unique and dynamic.

History of the Crystal Method

The crystal method was developed by an American scientist named ***Alistair Cockburn*** *who worked at IBM. He decided on team collaboration and communication.*

Traits of Cockburn's Crystal method were:

Human-powered *- the project should be flexible and people involved in preferred work.****Adaptive*** *- approaches can be changed anytime to meet the team's specific needs.****Ultra-light*** *- This methodology doesn't require much documentation.*

Properties of Crystal Agile Framework

Frequent Delivery: *It allows you regularly deliver the products and test code to real users.* ***Osmotic Communication****: It allows information to flow in between members of a team as in osmosis.****Personal Safety:*** *Team members should feel safe to discuss ideas openly without any fear.****Focus:*** *Each member of the team knows what to do, which enables them to focus their attention.* ***Easy access to expert users:*** *It enhances team communication with users and gets regular feedback from real users.****Technical tooling:*** *Specific technical tools are used by the software development team during testing, management, and configuration. These tools make the team to identify any error within less time.****Continuous learning:*** *Continuous learning enables the team members to acquire new skills and knowledge, and apply them in their work.****Teamwork:*** *The importance of teamwork, promoting collaboration and mutual support among*

*team members.**Time boxing:** It adopts time boxing to manage project deadlines and deliver the project in time.**Incremental development:** It promotes incremental development and adapt to changes.*

Crystal Methodology Operation

Crystal family consists of many variants like Crystal Clear, Crystal Yellow, Crystal Red, Crystal Sapphire, Crystal Red, Crystal Orange Web, and Crystal Diamond.

***Crystal Clear-** The team consists of only 1-6 members that is suitable for short-term projects where members work out in a single workspace.**Crystal Yellow-** It has a small team size of 7-20 members, where feedback is taken from Real Users. This variant involves automated testing and reduces the use of too much documentation.**Crystal Orange-** It has a team size of 21-40 members, where the team is split according to their functional skills. Here the project generally lasts for 1-2 years and the release is required every 3 to 4 months.**Crystal Orange Web-** It has also a team size of 21-40 members were the projects that have a continually evolving code base that is being used by the public. **Crystal Red-** The software development is led by 40-80 members where the teams can be formed and divided according to requirements.**Crystal Maroon-** It involves large-sized projects where the team size is 80-200 members and methods are different and as per the requirement of the software.**Crystal Diamond & Sapphire-** This variant is used in large projects where there is a potential risk to human life.*

Benefits

**** Collaboration**: Facilitate and enhance team communication and accountability.*

***Adaptability**: The adaptive approach lets the team respond well to the demanding requirements.**Higher quality:** Resulting in a higher quality product.**Improved customer satisfaction:** The team delivers products to meet customer needs.*

Drawbacks

***Ambiguity**: A lack of pre-defined plans may lead to confusion and loss of focus.**Disorganization**: Lack of structure may slow down inexperienced teams.**Uncertainty**: Not clear on how a remote team can share knowledge informally.*

3. Explain the life cycle of Extreme Programming(XP).

Answer:

EXTREME PROGRAMMING (XP)

It highlights on rapid, iterative development and deployment.

Life Cycle of Extreme Programming (XP)

The Extreme Programming Life Cycle consists of five phases:

Life Cycle of Extreme Programming (XP)

Planning:

The first stage is planning. During this phase, clients define their needs in short descriptions known as user stories.

It improves code quality and knowledge sharing.

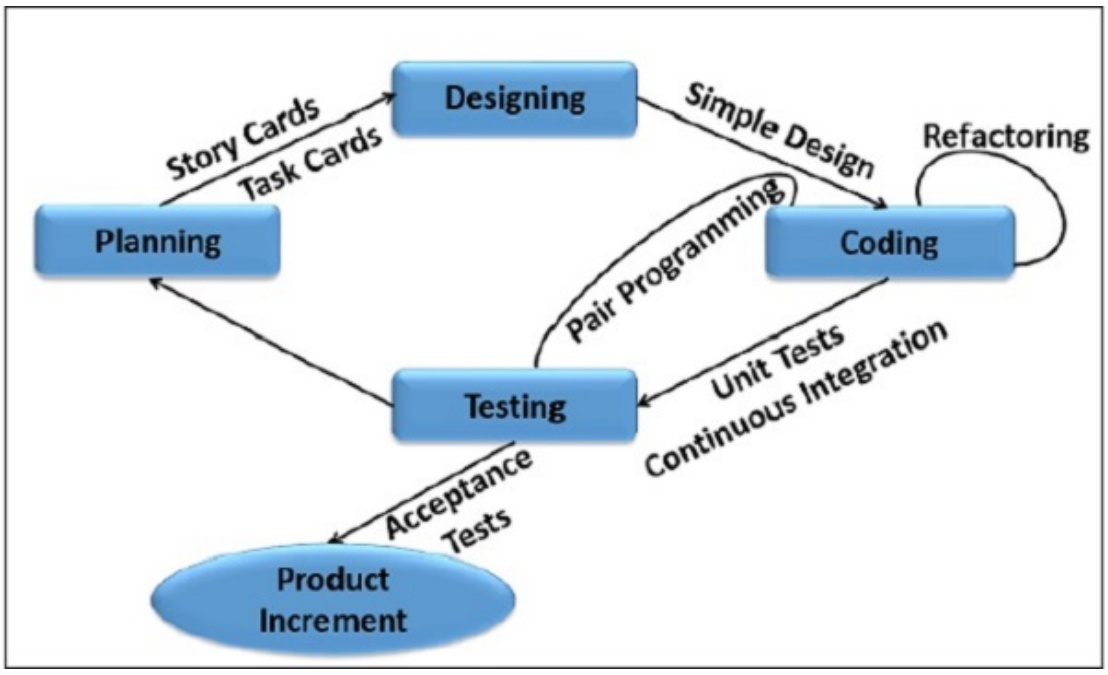
Testing:

Unit tests are automated to check if specific features work correctly.

·Acceptance tests are conducted by customers to ensure the overall system meets initial requirements.

·This continuous testing ensures the software's quality and alignment with customer needs.

***Listening:** In the listening phase regular feedback from customers to ensure the product meets their needs and to adapt to any changes.*



4. Explain Adaptive Software Development in Agile processes.

Answer: ***ADAPTIVE SOFTWARE DEVELOPMENT***

Adaptive Software Development *is a method to build complex software and systems.*

ASD focuses on human collaboration and self-organization.

Adaptive software development (ASD) is a software development process that grew out of the work by Jim Highsmith and Sam Bayer on rapid application development (RAD).

Characteristics of ASD Process

1. ***Adaptability:*** *It focuses on high-level adaptive planning instead of fixed plans.*

2. ***Collaborative Environment:*** *A strong focus is on collaboration within the team.*

3. ***Continuous Learning:*** *It highlights learning from each iteration and using feedback to improve the process and the product.*

*4.****Iterative Development:*** *The development is broken into small, manageable increments called working software.*

Responsive to Change: *It allows the project to quickly adapt to the changes in requirements, technology, and market conditions.****Risk Management:*** *Management of the risks with the flexibility.****Emphasizes Quality:*** *Regularly improve the code to maintain quality and adaptability.****Empowerment and Ownership:*** *It allows teams to make decisions and take ownership of their work. It also encourages decision-making at the team level to promote responsiveness.*

Phases of ASD Process

ASD life cycle incorporates three phases namely:

Speculation Collaboration Learning

These are explained below:

1. Speculation

During this phase project is initiated and planning is conducted.

It is the difficult part of ASD as it needs the workers to be motivated.

It highlights communication and teamwork.

It highlights individualism as individual creativity plays a major role in creative thinking.

People working together must trust each other to

Criticize without hatred Assist without anger, Work as hard as possible, Possession of skill set, Communicate problems to find effective solutions.

3. Learning

Learning helps the workers to increase their level of understanding of the project.

The learning process is in 3 ways:

Focus groups Technical reviews Project postmortem

Strengths of ASD

Customer Focus: *Regular involvement of customers to make sure that the product meets their needs and expectations.****Continuous Delivery:*** *Delivering functional software in small, allows for early and continuous delivery of value.****Early Problem Detection:*** *Regular testing help to identify and resolve issues early.****Strong Teamwork:*** *Highlights teamwork among the team members.* ***Empowered Teams:*** *Teams are given the autonomy to make decisions, leading to higher motivation.*

Weakness of ASD

Requires High Level of Skill: *Itrequires a team with a high-level of skill, experience, and familiarity with the agile principles.****Changing Requirements:*** *The frequent changes of requirements result in scope creep if it is not managed properly.****Customer Availability:*** *It can be difficult to achieve in cases when the customer is not available.****Uncertain Project Outcomes:*** *It difficult to predict the outcome of the project.*

Unit 4

1. Explain the stages in Institutional Knowledge Evolution Cycle.

Answer:

Institutional knowledge evolution cycle

The institutional knowledge evolution cycle in Agile software engineering, outlines how organizations develop, manage, and leverage collective knowledge to improve their software development processes. The cycle is typically described in five key stages:

Knowledge Development*: This initial stage focuses on the creation of new knowledge within the organization, often through learning, innovation, creativity, and importing experiences from external sources.****Knowledge Acquisition****: In this stage, the organization captures and retains information from various sources for future use and further processing.****Knowledge Refinement****: This involves organizing, transforming, and structuring the acquired knowledge, potentially incorporating it into written documentation or knowledge bases to ensure its usability.****Knowledge Distribution****: Knowledge is then disseminated throughout the organization through various channels, including education, training programs, automated knowledge-based systems, expert networks, and shared experiences.****Knowledge Deployment and Leveraging****: Finally, the acquired and distributed knowledge is applied in real-world situations, fostering further learning and innovation within the organization.*

2. Explain the different levels in Story card Maturity Model. (SMM)

Answer:

Story Card Maturity Model (SMM)

The Story Card Maturity Model (SMM) is a conceptual framework used in Agile software development to assess and improve the quality and readiness of user stories (or story cards) before and during a sprint. It helps teams ensure that each story card is clear, complete, and ready for implementation. Like other maturity models, SMM defines progressive levels of maturity.

Typical Levels of the Story Card Maturity Model

Here's a 5-level version commonly used across Agile teams:

Level 1 – Initial (Raw Idea)

***Characteristics**: High-level concept only. No acceptance criteria. Not estimated or prioritized.**Status**: Not ready for development.**Example**: "Improve login performance."*

Level 2 – Formulated

***Characteristics**: Story is written in a proper user story format (e.g., As a user, I want...). Initial discussion happened, but details are vague. Some business value identified.**Status**: Needs further refinement.*

Level 3 – Refined

***Characteristics**: Clear acceptance criteria are defined. Dependencies are identified. Estimation is possible. Small enough to be completed in one sprint.**Status**: Nearly "Ready."*

Level 4 – Ready for Sprint

***Characteristics**: Meets the **Definition of Ready (DoR)**. All team members understand the scope. Test cases are considered or defined. Prioritized in the backlog.**Status**: Can be pulled into sprint planning.*

Level 5 – Best Practices / Exemplary

***Characteristics**: Includes mockups, technical notes, and non-functional requirements. Linked to epics or features. Reflected in documentation tools (e.g., JIRA, Azure DevOps). Reviewed and approved by stakeholders or Product Owner.**Status**: Model story card; can be used as a reference.*

Usage of the SMM

Agile Coaches or Scrum Masters can use this model to train new team members. Product Owners use it to ensure backlog grooming/refinement leads to actionable stories. Development Teams benefit from consistently high-quality, ready-to-start work.

3. Explain the characteristics, benefits and challenges of Agile Information Systems.

Answer:

Agile Information Systems

Agile Information Systems (AIS) refer to information systems designed, developed, and maintained using agile methodologies to deliver flexible, adaptive, and user-focused solutions. These systems prioritize rapid development, iterative improvements, and responsiveness to changing requirements over rigid, traditional approaches like the waterfall model.

Characteristics of Agile Information Systems

*1. **Iterative Development**: AIS are built through short, iterative cycles (sprints) allowing continuous feedback and incremental improvements.*

*2. **User-Centric Design**: Emphasis on collaboration with stakeholders to ensure the system meets user needs and expectations.*

*3. **Flexibility and Adaptability**: Systems are designed to accommodate changing requirements, technologies, and business environments.*

*4. **Collaboration:** Cross-functional teams, including developers, testers, and business users, work closely to deliver functional components.*

*5. **Continuous Delivery**: Frequent releases of working software ensure quicker value delivery and the ability to refine features over time.*

*6.**Feedback Loops:** Regular feedback from users and stakeholders drives improvements and ensures alignment with goals.*

Benefits of Agile Information Systems

***Faster Time-to-Market:** Incremental releases allow usable features to be deployed sooner.*

***Improved Quality**: Continuous testing and feedback reduce defects and enhance reliability.*

***Customer Satisfaction**: Close collaboration ensures the system aligns with user needs.*

***Cost Efficiency:** Early detection of issues and iterative adjustments reduce rework costs.*

***Adaptability**: Systems can evolve with emerging technologies or market demands.*

Challenges in Agile Information Systems

***Cultural Resistance**: Organizations accustomed to traditional methods may struggle with agile's collaborative, iterative nature.*

***Scalability:** Agile practices can be harder to implement in large, distributed teams or complex systems.*

***Scope Creep:** Frequent changes can lead to uncontrolled expansion of project scope without clear prioritization.*

***Documentation:** Agile's focus on working software over documentation can create gaps in system knowledge.*

4. Explain the mapping of Earl_s schools to Agile Product Development.

Answer:

Mapping Earl's Schools to Agile Product Development

School	Relevance in Agile	Examples in Agile Context
Systems	*Agile teams can use tools (e.g., , Jira Confluence) to capture retrospectives, user stories, and technical knowledge.*	*Wikis for product documentation; knowledge bases for reusable patterns.*
Cartographic	*Helps teams quickly find domain experts or historical product knowledge in large organizations.*	*Expert directories or team skill matrices.*
Engineering	*KM is embedded in Agile ceremonies (retrospectives, sprint reviews) and DevOps pipelines.*	*Retrospectives driving continuous improvement; Continuous Integration /Continuous Delivery capturing learnings.*
Commercial	*Relevant when Agile is tied to product innovation, IP generation, or monetizing team knowledge.*	*Patenting features or leveraging product learnings as consulting offerings.*
Organizational	*Core to Agile – emphasizes collaboration, shared learning, team empowerment.*	*Communities of Practice, cross-functional teams, Scrum of Scrums.*
Spatial	*Agile thrives in shared spaces – physical or virtual – that encourage collaboration.*	*War rooms, daily stand-ups, virtual whiteboards (e.g., Miro, Mural).*
Strategic	*Agile KM should align with product and business strategy (e.g., using Objectives and Key Results or product roadmaps to drive knowledge development).*	*KM initiatives tied to Agile transformation or organizational learning goals.*

Unit 5

1. Explain the purpose and categories of Agile metrics.

Answer:

Agile Metrics

***Agile Metrics** are measurements used to track, assess, and improve the performance, quality, and value delivery of Agile teams. They focus on **outcomes** (value delivered) and **process health** rather than just raw productivity.*

Purpose of Agile Metrics

*Monitor **progress** toward goals. Identify **bottlenecks** and improvement areas. Ensure the team is delivering **business value**, not just output. Support **predictable delivery**.*

Categories of Agile Metrics

A. Productivity & Delivery Metrics

Measure how much work the team is completing.

***Velocity** – Amount of work (story points) completed per sprint.**Throughput** – Number of work items delivered per sprint.**Lead Time** – Time from request to delivery.**Cycle Time** – Time from starting a work item to finishing it.*

B. Predictability Metrics

Show how consistent and reliable the team's delivery is.

***Sprint Burndown Chart** – Tracks remaining work in the sprint.**Release Burndown Chart** – Tracks progress toward release goals.**Commitment Reliability** – Percentage of committed work completed.*

C. Quality Metrics

Ensure the product meets standards and user expectations.

***Defect Density** – Number of defects per size of work delivered.**Escaped Defects** – Bugs found after release.**Test Coverage** – Percentage of code covered by automated tests.*

D. Value Metrics

Measure actual business or customer value.

***Customer Satisfaction (CSAT)** – Direct feedback from users.**Net Promoter Score (NPS)** – Likelihood customers will recommend the product.**Business Value Delivered** – Features ranked by value completed.*

E. Team Health Metrics

Reflect collaboration, morale, and Agile maturity.

***Team Happiness Index** – Survey on team morale.**Collaboration Metrics** – Cross-functional engagement and communication quality.*

2. Define Requirement Elicitation and Explain Requirement Elicitation methods.

Answer:

Requirements Elicitation

The process of investigating and learning about a system's requirements from users, clients, and other stakeholders is known as requirements elicitation. Requirements elicitation in software engineering is most difficult, most error-prone and most communication-intensive software development

Requirement Elicitation Methods

***Interviews**:*

Direct, one-on-one or small group conversations with stakeholders to gather detailed information about their needs and expectations.

***Surveys/Questionnaires**:*

Distributing predefined questions to a large number of stakeholders to collect information efficiently.

***User Observation (Ethnography)**:*

Observing users in their natural work environment to understand their processes and identify unspoken needs or challenges.

***Workshops/Focus Groups**:*

Bringing together multiple stakeholders in a collaborative environment to discuss requirements, generate ideas, and reach consensus.

***Brainstorming**:*

A group creativity technique for generating a wide range of ideas and requirements in a short period.

***Prototyping**:*

Developing a working model or mock-up of the software to gather feedback from stakeholders and validate requirements.

***Use Cases**:*

Describing system interactions from an end-user's perspective, showing how they would use the system to achieve specific goals.

***Background Reading/Analysis**:*

Investigating existing documents, systems, or domain information to understand the context and constraints for the new system.

***Role-Playing/Scripted Walkthroughs**:*

Participants act out specific roles to explore system functionality and identify misunderstandings or missing information.

***Storyboarding/Class Responsibility- Collaboration(CRC) Cards**:*

Visual techniques used to depict user journeys or system responsibilities, helping to clarify requirements.

3. Explain Agile Requirements Prioritization and its techniques.

Answer:

Agile Requirements Prioritization

Agile Requirements Prioritization is the process of deciding which requirements (features, user stories, or backlog items) should be developed first so that the team delivers the highest business value as early as possible. In Agile, prioritization is continuous and adapts to feedback, market changes, and technical insights.

Need for Prioritization in Agile

*Delivers **maximum value early**. Makes the most of **limited time and budget**. Adapts to **changing requirements** without chaos. Helps stakeholders make **trade-offs** between features.*

Responsible Person

***Product Owner** (Scrum) or equivalent role owns backlog prioritization. Inputs come from **stakeholders, users, business analysts, and the development team**.*

Common Agile Prioritization Techniques

Technique	***How It Works***	***When to Use***
MoSCoW Method	*Categorize into Must have, Should have, Could have, Won't have (this time).*	*Small/medium projects needing quick classification.*
Kano Model	*Classify features as Basic, Performance, or Excitement to maximize customer satisfaction.*	*When customer experience is critical.*
Business Value Scoring	*Assign numerical values to features based on business benefit.*	*Projects with measurable ROI goals.*
Cost of Delay (CoD)	*Prioritize items that lose the most value if delayed.*	*Time-sensitive products or market-driven deadlines.*
WSJF (Weighted Shortest Job First)	*Prioritize by dividing Cost of Delay by Job Size (effort).*	*Large-scale projects (SAFe, portfolio level).*
Story Mapping	*Arrange stories in workflow order, then prioritize vertically by importance.*	*When visualizing the user journey is important.*
Relative Ranking / 100-Point Method	*Stakeholders distribute points across features.*	*When multiple decision-makers must agree.*

Agile Prioritization Workflow

Gather requirements *(user stories, epics, technical tasks).****Define prioritization criteria*** *(business value, urgency, risk, dependencies).****Choose a prioritization method*** *(MoSCoW, WSJF, etc.).****Prioritize the product backlog****.****Review and adjust*** *regularly during backlog refinement.****Select top-priority items*** *for the next sprint.*

4. Explain the Agile strategies for handling unstable requirements and its advantages.

Answer:

Managing Unstable Requirements

In many projects, requirements change because:

Market changes (competitors, trends, regulations). Stakeholder feedback after seeing working software. Technical discovery during development. Evolving user needs.

Agile Strategies for Handling Unstable Requirements

A. Iterative & Incremental Delivery

*Build in short cycles (****sprints****) so requirements can be revisited regularly. Changes can be applied at the* ***start of the next iteration*** *instead of disrupting ongoing work.*

B. Product Backlog Management

Keep all requirements in a ***prioritized backlog****. Unstable or unclear items stay low-priority until they stabilize. Refinements sessions help clarify them over time.*

C. Stakeholder Collaboration

Involve customers ***every sprint*** *(reviews, demos, feedback loops). This early feedback prevents large-scale rework later.*

D. Just-in-Time Detailing

Avoid defining every detail too early. Flesh out requirements ***only when development is about to begin****.*

E. User Stories & Acceptance Criteria

Break big, vague requirements into ***smaller, testable user stories****. This makes even unstable requirements easier to manage and test.*

F. Change-Friendly Planning

Use ***rolling-wave planning*** *instead of fixed, long-term plans. Accept that later requirements will change and budget time for it.*

Benefits of Agile for Unstable Requirements

Lower rework costs *(because work is done in small increments).****Better alignment*** *with current business priorities.****Higher customer satisfaction*** *because the delivered features reflect up-to-date needs.*